

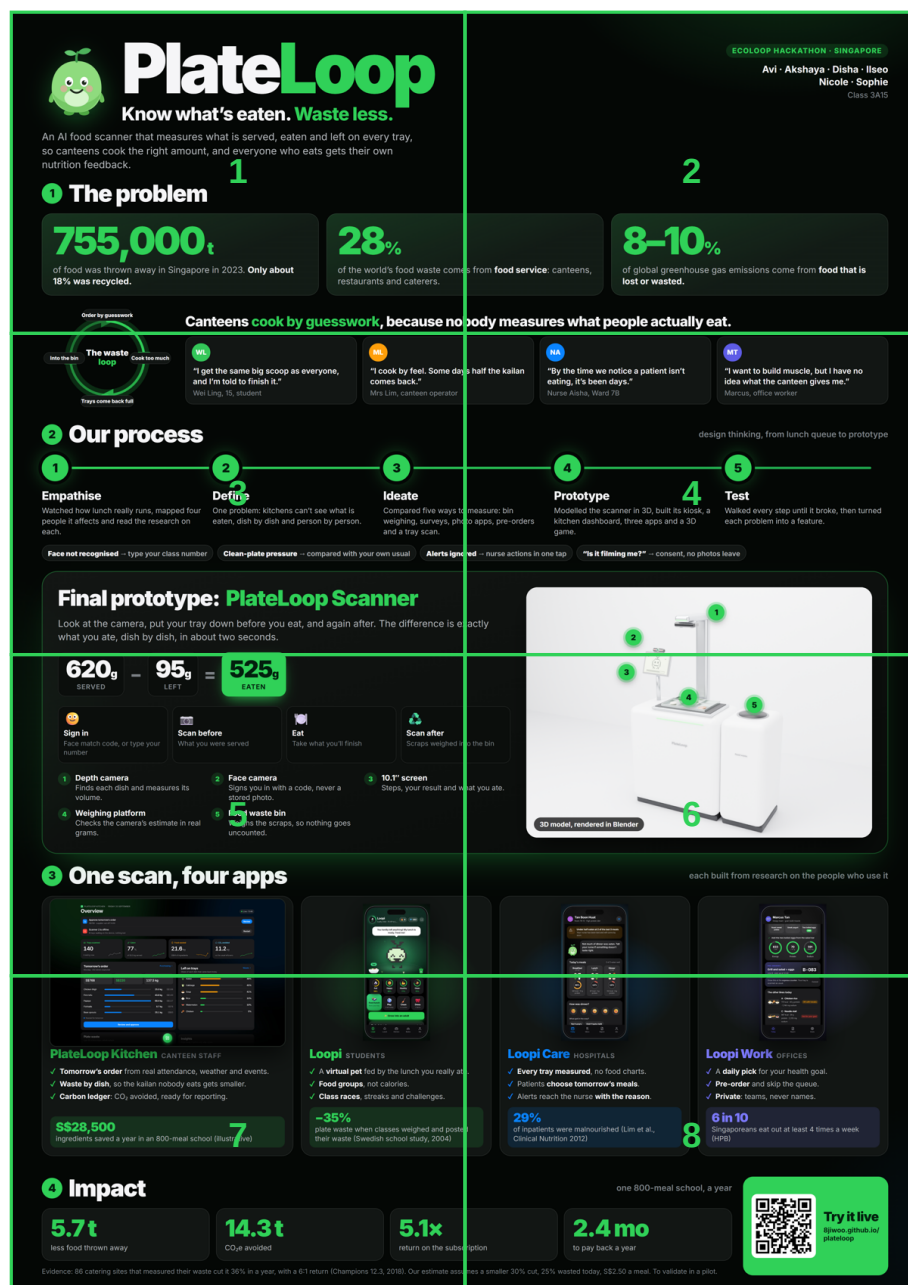
PlateLoop poster: A1 from 8 A4 sheets

Print every sheet on A4 at 100% (Actual size), landscape, with borderless printing on.

Each sheet is exactly one eighth of the A1 poster, edge to edge, so there is nothing to trim.

Lay them out as numbered below (2 across, 4 down) and tape them together from the back.

If your printer cannot print borderless, use the home-printer file instead: it has margins and overlaps.





PlateLoop

Know what's eaten. Waste less.

An AI food scanner that measures what is served, eaten and left on every tray, so canteens cook the right amount, and everyone who eats gets their own nutrition feedback.

1 The problem

755,000_t

of food was thrown away in Singapore in 2023. **Only about 18% was recycled.**

28%

of the world's food waste comes from restaurants and caterers.

Order by guesswork

Canteens **cook by guesswork**, because not

Avi · Akshaya · Disha · Ilseo
Nicole · Sophie

Class 3A15

from **food service**: canteens,

8–10%

of global greenhouse gas emissions come from **food that is lost or wasted**.

body measures what people actually eat.



WL

"I get the same big scoop as everyone, and I'm told to finish it."

Wei Ling, 15, student

ML

"I cook by feel. Some days it comes back."

Mrs Lim, canteen operator

2 Our process

1

Empathise

Watched how lunch really runs, mapped four people it affects and read the research on each.

Face not recognised → type your class number

2

Define

One problem: kitchens can't see what is eaten, dish by dish and person by person.

Clean-plate pressure → compared with your own usual

3

Ideate

Compared five ways to track food: weighing, surveys, photos, and a tray scan.

Alerts ignored

Final prototype: **PlateLoop Scanner**

Look at the camera, put your tray down before you eat, and again after. The difference is exactly what you ate, dish by dish, in about two seconds.

s half the kailan

NA

"By the time we notice a patient isn't eating, it's been days."

Nurse Aisha, Ward 7B

MT

"I want to build muscle, but I have no idea what the canteen gives me."

Marcus, office worker

design thinking, from lunch queue to prototype

4

Prototype

measure: bin
to apps, pre-orders

Modelled the scanner in 3D, built its kiosk, a kitchen dashboard, three apps and a 3D game.

5

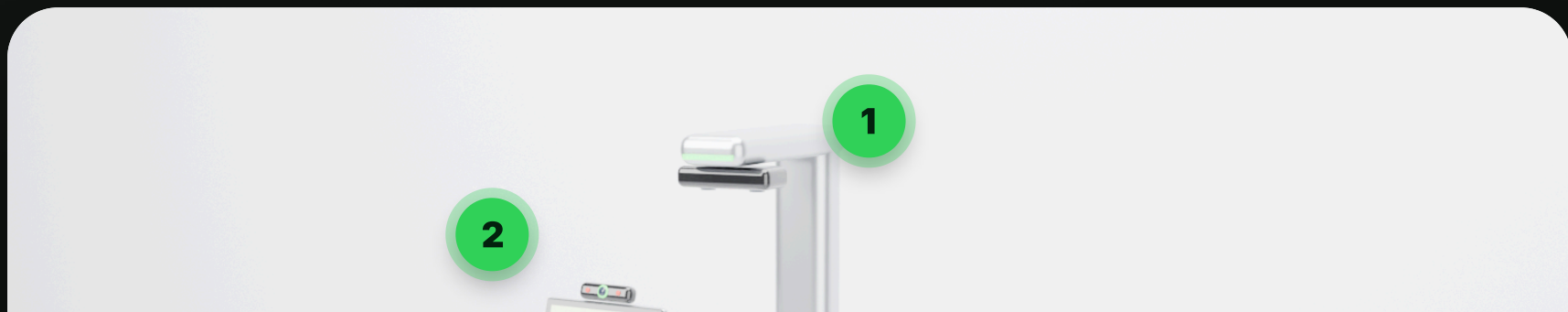
Test

Walked every step until it broke, then turned each problem into a feature.

red → nurse actions in one tap

"Is it filming me?" → consent, no photos leave

xactly



620_g
SERVED

—

95_g
LEFT

=

525_g
EATEN



Sign in

Face match code, or type your number



Scan before

What you were served



Eat

Take what you'll finish



Scan after

Scraps weighed in

1

Depth camera

Finds each dish and measures its volume.

2

Face camera

Signs you in with a code, never a stored photo.

3

10.1" screen

Steps, your result and w

4

Weighing platform

Checks the camera's estimate in real grams.

5

Food waste bin

Weighs the scraps, so nothing goes uncoun

3 One scan, four apps

PLATELOOP KITCHEN · FRIDAY 25 SEPTEMBER

Overview

Live · 11:48

Approve tomorrow's order

\$5768 · supplier cut-off 14:00

Review

Scanner 2 is offline

6 trays waiting on the device, nothing lost

Restart

Trays scanned

140

0 eating now

Eaten

77%

of 93.5 kg served

Food wasted

21.6 kg

\$84 of ingredients

CO₂ avoided

11.2 kg

vs the usual leftovers

Tomorrow's order

Monday · 812 diners expected

Purchasing >

Left on trays

Share of each dish that came back today

Waste >

4 Loopi

Leafy child · 73 XP to L...

4 162

You hardly left anything! My lunch is ready. Feed me!



Tap!

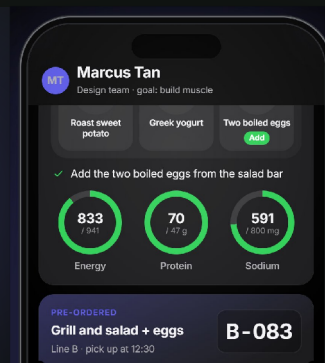
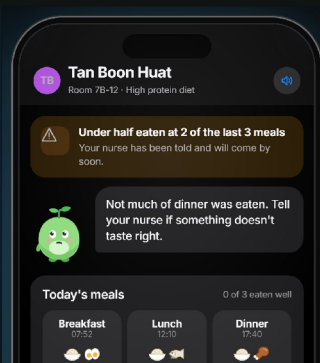
to the bin

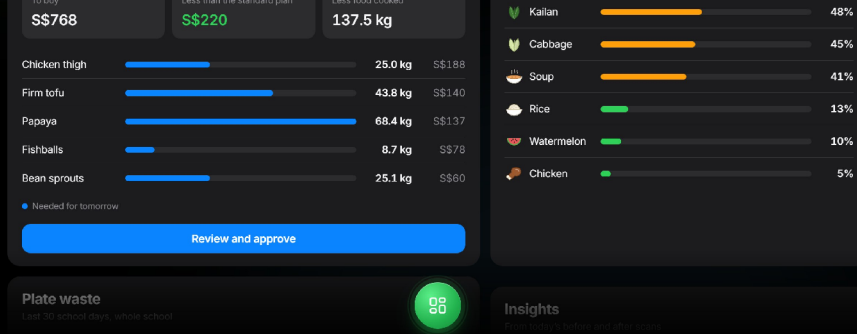
what you ate.



3D model, rendered in Blender

each built from research on the people who use it



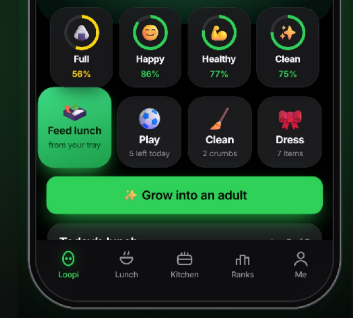


PlateLoop Kitchen CANTEEN STAFF

- ✓ **Tomorrow's order** from real attendance, weather and events.
- ✓ **Waste by dish**, so the kailan nobody eats gets smaller.
- ✓ **Carbon ledger**: CO₂ avoided, ready for reporting.

\$28,500

ingredients saved a year in an 800-meal school (illustrative)



Loopi STUDENTS

- ✓ A **virtual pet** fed by the lunch you really ate
- ✓ **Food groups**, not calories.
- ✓ **Class races**, streaks and challenges.

-35%

plate waste when classes weighed and posted their waste (Swedish school study, 2004)

4 Impact

5.7t

less food thrown away

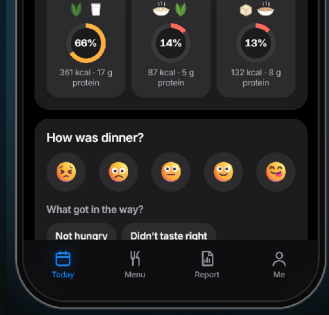
14.3t

CO₂e avoided

5.1x

return on the subscri

Evidence: 86 catering sites that measured their waste cut it 36% in a year, with a 6:1 return (Champions 12.3, 2018). Our estimate assum

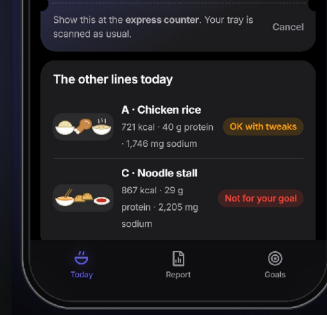


Loopi Care HOSPITALS

- ✓ **Every tray measured**, no food charts.
- ✓ Patients **choose tomorrow's meals**.
- ✓ Alerts reach the nurse **with the reason**.

29%

of inpatients were malnourished (Lim et al., Clinical Nutrition 2012)



Loopi Work OFFICES

- ✓ A **daily pick** for your health goal.
- ✓ **Pre-order** and skip the queue.
- ✓ **Private**: teams, never names.

6 in 10

Singaporeans eat out at least 4 times a week (HPB)

one 800-meal school, a year

2.4 mo

to pay back a year



Try it live

[8jiwoo.github.io/
plateloop](https://8jiwoo.github.io/plateloop)

cription

times a smaller 30% cut, 25% wasted today, S\$2.50 a meal. To validate in a pilot.